

SEMESTER S8

CLOUD COMPUTING AND APPLICATIONS

Course Code	OEERT 834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Operating System and Networking	Course Type	Theory

Course Objectives:

1. Understand the fundamentals and architecture of cloud computing, including delivery and deployment models.
2. Develop knowledge and skills in cloud security, SLA management, and risk assessment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cloud Computing Fundamentals: What is Cloud Computing, Essential Characteristics, Architectural Influences Cloud delivery models, The SPI Framework, Cloud Software as a Service (SaaS), Cloud Platform as a Service (PaaS), Cloud Infrastructure as a Service (IaaS), Cloud deployment models, Public Clouds, Community Clouds, Private clouds, Hybrid clouds.	9
2	Understanding cloud architecture- Exploring cloud computing stack. Using virtualization technologies, Load balancing and virtualization, Understanding hypervisors, Understanding machine imaging. Cloud storage- Cloud storage providers, Cloud Computing with the Titans– Google, Amazon. Accessing the cloud: Platforms, Web applications, Web APIs.	9
3	Virtual Machines and Containers, Serverless Computing, Using and Managing Containers:-Container Basics, Docker and the Hub. Agents and Microservices: Microservices and Container Resource Managers, Managing Identity in a swarm, A simple Microservices Example, Amazon EC2 Container Service, Google’s Kubernetes.	9

4	SLA Management in Cloud Computing-Types of SLA, Life cycle of SLA, SLA management in cloud. Cloud Information Security Objectives, Cloud Security Services, Relevant Cloud Security Design Principles, Secure Cloud Software Requirements. Privacy and Compliance Risks, Threats to Infrastructure Data and Access Control, Cloud Service Provider Risks. Cloud computing Security Architecture- architecture considerations, identity management and access control	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the principles of cloud computing.	K2
CO2	Explain the technologies used in cloud computing and virtualization.	K2
CO3	Describe cloud computing and microservices.	K2
CO4	Describe cloud management and cloud security features.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	-	-	-	-	-	-	-	2
CO2	3	2	-	-	-	-	-	-	-	-	-	2
CO3	3	2	-	-	-	-	-	-	-	-	-	2
CO4	3	2	-	-	-	-	-	-	-	-	-	2
CO5	3	2	-	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cloud Security -A Comprehensive Guide to Secure Cloud Computing	Ronald L Krutz and Russell Dean Vines	Wiley Publishing, Inc.	2010
2	Cloud Computing Bible	Barrie Sosinsky	Wiley Publishing	2011
3	Cloud Computing: A Practical Approach	Anthony T. Velte Toby J. Velte, Robert Elsenpeter	The McGraw-Hill	2010
4	Cloud Computing for Science and Engineering https://cloud4scieng.org/chapters/	Ian Foster and Dennis B.Gannon	The MIT Press	2017
5	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg and Andrzej M. Goscinski	Wiley Publishing	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Distributed and Cloud Computing: From parallel processing to Internet of Things	Kai Hwang, Geoffrey C. Fox, Jack K. Dongarra	Morgan Kaufmann	2013
2	Getting Started with Kubernetes: 2nd Edition	Jonathan Baier	Packt publishers	2 nd , 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://en.wikipedia.org/wiki/Cloud_computing https://nptel.ac.in/courses/106105167
2	https://nptel.ac.in/courses/106105167
3	https://nptel.ac.in/courses/106105167
4	https://nptel.ac.in/courses/106105167